

A. Introduction

1. **Title:** Cyber Security — Incident Reporting and Response Planning
2. **Number:** RRC-CIP-008-1
3. **Purpose:** To mitigate the risk to the reliable operation of the Bulk Electric System (BES) as the result of a Cyber Security Incident by specifying incident response requirements.
4. **Applicability:**
 - 4.1.1. **Balancing Authority/Load Balancing Authority**
 - 4.1.2. **Generator Operator**
 - 4.1.3. **Generator Owner**
 - 4.1.4. **Transmission Operator**
 - 4.1.5. **Transmission Owner**
 - 4.1.6. **Distribution Provider:** that owns one or more of the following Facilities, systems, and equipment for the protection or restoration of the BES:
 - 4.1.6.1. Each underfrequency load shedding (UFLS) or undervoltage loadshedding (UVLS) system that:
 - 4.1.6.1.1. is part of a Load shedding program that is subject to one or more requirements in a Railbelt Reliability Standard; and
 - 4.1.6.1.2. performs automatic Load shedding under a common control system owned by the Responsible Entity, without human operator initiation, that the loss, compromise, or misuse could have an adverse impact on the reliable operation of the BES.
 - 4.1.6.2. Remedial Action Scheme (RAS) where the RAS is subject to one or more requirements in a Railbelt Reliability Council Standard.
 - 4.1.6.3. Each Protection System (excluding UFLS and UVLS) that applies to Transmission where the Protection System is subject to one or more requirements in a Railbelt Reliability Council Standard.
 - 4.1.6.4. Each Cranking Path and group of Elements meeting the initial switching requirements from a Blackstart Resource up to and including the first interconnection point of the starting station service of the next generation unit(s) to be started.
- 4.2. **Facilities:** For the purpose of the requirements contained herein, the following Facilities, systems, and equipment owned by each Responsible Entity in 4.1 above are those to which these requirements are applicable. For requirements in this standard where a specific type of Facilities, system, or equipment or subset of Facilities, systems, and equipment are applicable, these are specified explicitly.
 - 4.2.1. **Distribution Provider:** One or more of the following Facilities, systems and equipment owned by the Distribution Provider for the protection or restoration of the BES:
 - 4.2.1.1. Each UFLS or UVLS System that:
 - 4.2.1.1.1. is part of a Load shedding program that is subject to one or more

requirements in a Railbelt Reliability Council Standard; and

- 4.2.1.1.2.** performs automatic Load shedding under a common control system owned by the Responsible Entity, without human operator initiation, that the loss, compromise, or misuse of could have adverse impact on the reliable operation of the BES.

4.2.1.2. Each RAS where the RAS is subject to one or more requirements in a Railbelt Reliability Council Standard.

4.2.1.3. Each Protection System (excluding UFLS and UVLS) that applies to Transmission where the Protection System is subject to one or more requirements in a Railbelt Reliability Council Standard.

4.2.1.4. Each Cranking Path and group of Elements meeting the initial switching requirements from a Blackstart Resource up to and including the first interconnection point of the starting station service of the next generation unit(s) to be started.

4.2.2. Responsible Entities listed in 4.1 other than Distribution Providers:

All BES Facilities.

4.2.3. Exemptions: The following are exempt from Standard **RRC-CIP-008-1**:

4.2.3.1. Cyber Systems at Facilities regulated by the Canadian Nuclear Safety Commission.

4.2.3.2. Cyber Assets associated with communication networks and data communication links between discrete Electronic Security Perimeters (ESP).

4.2.3.3. Cyber Systems, associated with communication networks and data communication links, between Cyber Systems, providing confidentiality and integrity of an ESP that extends to one or more geographic locations.

4.2.3.4. The systems, structures, and components that are regulated by the Nuclear Regulatory Commission under a cyber security plan pursuant to 10 C.F.R. Section 73.54.

4.2.3.5. For Distribution Providers, the systems and equipment that are not included in section 4.2.1 above.

4.2.3.6. Responsible Entities that identify that they have no BES Cyber Systems categorized as high impact or medium impact according to the CIP-002 identification and categorization processes.

4.3. “Applicable Systems”: Each table has an “Applicable Systems” column to define the scope of systems to which a specific Requirement Part applies.

5. Effective Dates: Shall become effective on the first day of the first calendar quarter that is twenty-four (24) months following adoption by the Approving Authority.

B. Requirements and Measures

- R1.** Each Responsible Entity shall document one or more Cyber Security Incident response plan(s) that collectively include each of the applicable requirement parts in RRC-CIP-008-1 *Table R1 – Cyber Security Incident Response Plan Specifications*. [Violation Risk Factor: Lower]
- M1.** Evidence must include each of the documented plan(s) that collectively include each of the applicable requirement parts in RRC-CIP-008-1 *Table R1 – Cyber Security Incident Response Plan Specifications*.

RRC-CIP-008-1 Table R1 – Cyber Security Incident Response Plan Specifications			
Part	Applicable Systems	Requirements	Measures
1.1	High impact BCS and their associated Electronic Access Control and Monitoring Systems (EACMS) Medium impact BCS and their associated EACMS Shared Cyber Infrastructure (SCI) supporting an Applicable System in this Part	One or more processes to identify, classify, and respond to Cyber Security Incidents.	Examples of evidence may include, but are not limited to, dated documentation of Cyber Security Incident response plan(s) that include the process(es) to identify, classify, and respond to Cyber Security Incidents.

RRC-CIP-008-1 Table R1 – Cyber Security Incident Response Plan Specifications			
Part	Applicable Systems	Requirements	Measures
1.2	High impact BCS and their associated EACMS Medium impact BCS and their associated EACMS SCI supporting an Applicable System in this Part	One or more processes: 1.2.1 That include criteria to evaluate and define attempts to compromise; 1.2.2 To determine if an identified Cyber Security Incident is: • A Reportable Cyber Security Incident; or • An attempt to compromise, as determined by applying the criteria from Part 1.2.1, one or more systems identified in the Applicable Systems column for this Part; and 1.2.3 To provide notification per Requirement R4.	Examples of evidence may include, but are not limited to, dated documentation of Cyber Security Incident response plan(s) that provide guidance or thresholds for determining which Cyber Security Incidents are also Reportable Cyber Security Incidents or a Cyber Security Incident that is determined to be an attempt to compromise a system identified in the Applicable Systems column including justification for attempt determination criteria and documented processes for notification.
1.3	High impact BCS and their associated EACMS Medium impact BCS and their associated EACMS SCI supporting an Applicable System in this Part	The roles and responsibilities of Cyber Security Incident response groups or individuals.	Examples of evidence may include, but are not limited to, dated Cyber Security Incident response process(es) or procedure(s) that define roles and responsibilities (e.g., monitoring, reporting, initiating, documenting, etc.) of Cyber Security Incident response groups or individuals.

RRC-CIP-008-1 Table R1 – Cyber Security Incident Response Plan Specifications			
Part	Applicable Systems	Requirements	Measures
1.4	High impact BCS and their associated EACMS Medium impact BCS and their associated EACMS SCI Supporting an Applicable System in this Part	Incident handling procedures for Cyber Security Incidents.	Examples of evidence may include, but are not limited to, dated Cyber Security Incident response process(es) or procedure(s) that address incident handling (e.g., containment, eradication, recovery/incident resolution).

- R2.** Each Responsible Entity shall implement each of its documented Cyber Security Incident response plans to collectively include each of the applicable requirement parts in RRC-CIP-008-1 *Table R2 – Cyber Security Incident Response Plan Implementation and Testing*.
[Violation Risk Factor: Lower]
- M2.** Evidence must include, but is not limited to, documentation that collectively demonstrates implementation of each of the applicable requirement parts in RRC-CIP-008-1 *Table R2 – Cyber Security Incident Response Plan Implementation and Testing*.

RRC-CIP-008-1 Table R2 – Cyber Security Incident Response Plan Implementation and Testing			
Part	Applicable Systems	Requirements	Measures
2.1	High impact BCS and their associated EACMS Medium impact BCS and their associated EACMS SCI supporting an Applicable System in this Part	Test each Cyber Security Incident response plan(s) at least once every 15 calendar months: • By responding to an actual Reportable Cyber Security Incident; • With a paper drill or tabletop exercise of a Reportable Cyber Security Incident; or • With an operational exercise of a Reportable Cyber Security Incident.	Examples of evidence may include, but are not limited to, dated evidence of a lessons-learned report that includes a summary of the test or a compilation of notes, logs, and communication resulting from the test. Types of exercises may include discussion or operations based exercises.
2.2	High impact BCS and their associated EACMS Medium impact BCS and their associated EACMS SCI supporting an Applicable System in this Part	Use the Cyber Security Incident response plan(s) under Requirement R1 when responding to a Reportable Cyber Security Incident, responding to a Cyber Security Incident that attempted to compromise a system identified in the Applicable Systems column for this Part, or performing an exercise of a Reportable Cyber Security Incident. Document deviations from the plan(s) taken during the response to the incident or exercise.	Examples of evidence may include, but are not limited to, incident reports, logs, and notes that were kept during the incident response process, and follow-up documentation that describes deviations taken from the plan during the incident response or exercise.

RRC-CIP-008-1 Table R2 – Cyber Security Incident Response Plan Implementation and Testing			
Part	Applicable Systems	Requirements	Measures
2.3	High impact BCS and their associated EACMS Medium impact BCS and their associated EACMS SCI supporting an Applicable System in this Part	Retain records related to Reportable Cyber Security Incidents and Cyber Security Incidents that attempted to compromise a system identified in the Applicable Systems column for this Part as per the Cyber Security Incident response plan(s) under Requirement R1.	Examples of evidence may include, but are not limited to, dated documentation, such as security logs, police reports, emails, response forms or checklists, forensic analysis results, restoration records, and post-incident review notes related to Reportable Cyber Security Incidents and a Cyber Security Incident that is determined to be an attempt to compromise a system identified in the Applicable Systems column.

- R3.** Each Responsible Entity shall maintain each of its Cyber Security Incident response plans according to each of the applicable requirement parts in RRC-CIP-008-1 *Table R3 – Cyber Security Incident Response Plan Review, Update, and Communication*. [Violation Risk Factor: Lower].
- M3.** Evidence must include, but is not limited to, documentation that collectively demonstrates maintenance of each Cyber Security Incident response plan according to the applicable requirement parts in RRC-CIP-008-1 *Table R3 – Cyber Security Incident Response Plan Review, Update, and Communication*.

RRC-CIP-008-1 Table R3 – Cyber Security Incident Response Plan Review, Update, and Communication			
Part	Applicable Systems	Requirements	Measures
3.1	High Impact BCS and their associated EACMS Medium impact BCS and their associated EACMS SCI supporting an Applicable System in this Part	No later than 90 calendar days after completion of a Cyber Security Incident response plan(s) test or actual Reportable Cyber Security Incident response: 3.1.1. Document any lessons learned or document the absence of any lessons learned; 3.1.2. Update the Cyber Security Incident response plan based on any documented lessons learned associated with the plan; and 3.1.3. Notify each person or group with a defined role in the Cyber Security Incident response plan of the updates to the Cyber Security Incident response plan based on any documented lessons learned.	Examples of evidence may include, but are not limited to, all of the following: 1. Dated documentation of post incident(s) review meeting notes or follow-up report showing lessons learned associated with the Cyber Security Incident response plan(s) test or actual Reportable Cyber Security Incident response or dated documentation stating there were no lessons learned; 2. Dated and revised Cyber Security Incident response plan showing any changes based on the lessons learned; and 3. Evidence of plan update distribution including, but not limited to: • Emails; • USPS or other mail service; • Electronic distribution system; or

			• Training sign-in sheets.
3.2	High impact BCS and their associated EACMS Medium impact BCS and their associated EACMS SCI supporting an Applicable System in this Part	No later than 60 calendar days after a change to the roles or responsibilities, Cyber Security Incident response groups or individuals, or technology that the Responsible Entity determines would impact the ability to execute the plan: 3.21. Update the Cyber Security Incident response plan(s); and 3.22. Notify each person or group with a defined role in the Cyber Security Incident response plan of the updates.	Examples of evidence may include, but are not limited to: 1. Dated and revised Cyber Security Incident response plan with changes to the roles or responsibilities, responders or technology; and 2. Evidence of plan update distribution including, but not limited to: • Emails; • USPS or other mail service; • Electronic distribution system; or • Training sign-in sheets.

- R4.** Each Responsible Entity shall notify the Electricity Information Sharing and Analysis Center (E-ISAC) and, if subject to the jurisdiction of the United States, the United States Cybersecurity & Infrastructure Security Agency (CISA), or their successors, of a Reportable Cyber Security Incident and a Cyber Security Incident that was an attempt to compromise, as determined by applying the criteria from Requirement R1, Part 1.2.1, a system identified in the Applicable Systems column, unless prohibited by law, in accordance with each of the applicable requirement parts in RRC-CIP-008-1 *Table R4 – Notifications and Reporting for Cyber Security Incidents*. [Violation Risk Factor: Lower] [Time Horizon: Operations Assessment].
- M4.** Evidence must include, but is not limited to, documentation that collectively demonstrates notification of each determined Reportable Cyber Security Incident and a Cyber Security Incident that was an attempt to compromise a system identified in the Applicable Systems column according to the applicable requirement parts in RRC-CIP-008-1 *Table R4 – Notifications and Reporting for Cyber Security Incidents*.

RRC-CIP-008-1 Table R4 – Notifications and Reporting for Cyber Security Incidents			
Part	Applicable Systems	Requirements	Measures
4.1	High impact BCS and their associated EACMS Medium impact BCS and their associated EACMS SCI supporting an Applicable System in this Part	Initial notifications and updates shall include the following attributes, at a minimum, to the extent known: 4.1.1 The functional impact; 4.1.2 The attack vector used; and 4.1.3 The level of intrusion that was achieved or attempted.	Examples of evidence may include, but are not limited to, dated documentation of initial notifications and updates to the E-ISAC and CISA, or their successors.

RRC-CIP-008-1 Table R4 – Notifications and Reporting for Cyber Security Incidents			
Part	Applicable Systems	Requirements	Measures
4.2	High impact BCS and their associated EACMS	After the Responsible Entity's determination made pursuant to documented process(es) in Requirement R1, Part 1.2, provide initial notification within the following timelines: • One hour after the determination of a Reportable Cyber Security Incident. • By the end of the next calendar day after determination that a Cyber Security Incident was an attempt to compromise a system identified in the Applicable Systems column for this Part.	Examples of evidence may include, but are not limited to, dated documentation of notices to the E-ISAC and CISA, or their successors.
	Medium impact BCS and their associated EACMS SCI supporting an Applicable System in this Part		
4.3	High impact BCS and their associated EACMS	Provide updates, if any, within seven calendar days of determination of new or changed attribute information required in Part 4.1.	Examples of evidence may include, but are not limited to, dated documentation of submissions to the E-ISAC and CISA, or their successors.
	Medium impact BCS and their associated EACMS SCI supporting an Applicable System in this Part		

C. Compliance

1. Compliance Monitoring Process

1.1 Compliance Enforcement Authority:

The RRC is the entity with the authority to monitor and enforce compliance with mandatory and enforceable Railbelt Reliability Council Standards.

1.2 Evidence Retention:

The following evidence retention periods identify the period of time an entity is required to retain specific evidence to demonstrate compliance. For instances where the evidence retention period specified below is shorter than the time since the last audit, the RRC may ask an entity to provide other evidence to show that it was compliant for the full time period since the last audit.

The Responsible Entity shall keep data or evidence to show compliance as identified below unless directed by the RRC to retain specific evidence for a longer period of time as part of an investigation:

- Each Responsible Entity shall retain evidence of each requirement in this standard for three calendar years.
- If a Responsible Entity is found non-compliant, it shall keep information related to the non-compliance until mitigation is complete and approved or the time specified above, whichever is longer.
- The RRC shall keep the last audit records and all requested and submitted subsequent audit records.

1.3 Compliance Monitoring and Enforcement Program:

Compliance Monitoring and Enforcement Program refers to the identification of the processes that will be used to evaluate data or information for the purpose of assessing performance or outcomes with the associated Railbelt Reliability Council Standards.

Table of Compliance Elements

R#	Violation Risk Factor (VRF)	Violation Severity Levels (RRC-CIP-008-1)			
		Lower VSL	Moderate VSL	High VSL	Severe VSL
R1	Lower	N/A	N/A	The Responsible Entity did not include the roles and responsibilities of Cyber Security Incident response groups or individuals. (R 1.3) OR The Responsible Entity did not include incident handling procedures for Cyber Security Incidents. (R 1.4) OR The Responsible Entity's plan did not include one or more processes to provide notification per Requirement R4. (R 1.2.3) OR The Responsible Entity's plan did not include one or more processes that include criteria to evaluate and define attempts to compromise. (R 1.2.1)	The Responsible Entity did not develop a Cyber Security Incident response plan with one or more processes to identify, classify, and respond to Cyber Security Incidents. (R 1.1) OR The Responsible Entity's plan did not include one or more processes to identify Reportable Cyber Security Incidents or a Cyber Security Incident that was an attempt to compromise, as determined by applying the criteria from R 1.2.1, a system identified in the Applicable Systems column for R 1.2. (R 1.2.2)

R#	Violation Risk Factor (VRF)	Violation Severity Levels (RRC-CIP-008-1)			
		Lower VSL	Moderate VSL	High VSL	Severe VSL
R2	Lower	The Responsible Entity did not test the Cyber Security Incident response plan(s) within 15 calendar months, not exceeding 18 calendar months between tests of the plan(s). (R 2.1)	The Responsible Entity did not test the Cyber Security Incident response plan(s) within 18 calendar months, not exceeding 21 calendar months between tests of the plan(s). (R 2.1)	The Responsible Entity did not test the Cyber Security Incident response plan(s) within 21 calendar months, not exceeding 24 calendar months between tests of the plan(s). (R 2.1) OR The Responsible Entity did not document deviations, if any, from the plan during a test or when a Reportable Cyber Security Incident or a Cyber Security Incident that was an attempt to compromise a system identified in the Applicable Systems column for R 2.2 occurs. (R 2.2)	The Responsible Entity did not test the Cyber Security Incident response plan(s) within 24 calendar months between tests of the plan(s). (R 2.1) OR The Responsible Entity did not retain relevant records related to Reportable Cyber Security Incidents or Cyber Security Incidents that were an attempt to compromise a system identified in the Applicable Systems column for R 2.3. (R 2.3)
R3	Lower	The Responsible Entity did not notify each person or group with a defined role in the Cyber Security Incident response plan of updates to the Cyber Security Incident response plan within greater than 90 but less than 105 calendar days of a test or actual incident response to a Reportable Cyber Security Incident. (R 3.1.3)	The Responsible Entity did not update the Cyber Security Incident response plan based on any documented lessons learned within 90 and less than 105 calendar days of a test or actual incident response to a Reportable Cyber Security Incident. (R 3.1.2) OR	The Responsible Entity neither documented lessons learned nor documented the absence of any lessons learned within 90 and less than 105 calendar days of a test or actual incident response to a Reportable Cyber Security Incident. (R 3.1.1) OR	The Responsible Entity neither documented lessons learned nor documented the absence of any lessons learned equal to or greater than 105 calendar days of a test or actual incident response to a Reportable Cyber Security Incident. (R 3.1.1)

R#	Violation Risk Factor (VRF)	Violation Severity Levels (RRC-CIP-008-1)			
		Lower VSL	Moderate VSL	High VSL	Severe VSL
			The Responsible Entity did not notify each person or group with a defined role in the Cyber Security Incident response plan of updates to the Cyber Security Incident response plan equal to or greater than 105 calendar days of a test or actual incident response to a Reportable Cyber Security Incident. (R 3.1.3) OR The Responsible Entity did not update the Cyber Security Incident response plan(s) or notified each person or group with a defined role within 60 and less than 75 calendar days of any of the following changes that the responsible entity determines would impact the ability to execute the plan: • Roles or responsibilities, or • Cyber Security Incident response groups or individuals, or • Technology changes. (R 3.2)	The Responsible Entity did not update the Cyber Security Incident response plan based on any documented lessons learned equal to or greater than 105 calendar days of a test or actual incident response to a Reportable Cyber Security Incident. (R 3.1.2) OR The Responsible Entity did not update the Cyber Security Incident response plan(s) or notified each person or group with a defined role equal to or greater than 75 calendar days of any of the following changes that the responsible entity determines would impact the ability to execute the plan: • Roles or responsibilities, or • Cyber Security Incident response groups or individuals, or • Technology changes. (R 3.2)	

R#	Violation Risk Factor (VRF)	Violation Severity Levels (RRC-CIP-008-1)			
		Lower VSL	Moderate VSL	High VSL	Severe VSL
R4	Lower	The Responsible Entity notified E-ISAC and CISA, or their successors, of a Cyber Security Incident that was an attempt to compromise a system identified in the “Applicable Systems” column for R 4.2 but failed to notify or update E-ISAC or CISA, or their successors, within the timelines pursuant to R 4.2 (R 4.2) OR The Responsible Entity notified E-ISAC and CISA, or their successors, of a Reportable Cyber Security Incident or a Cyber Security Incident that was an attempt to compromise a system identified in the “Applicable Systems” column for R 4.3 but failed to report on one or more of the attributes within 7 days after determination of the attribute(s) not reported pursuant to R 4.1. (R 4.3) OR	The Responsible Entity failed to notify E-ISAC or CISA, or their successors, of a Cyber Security Incident that was an attempt to compromise, as determined by applying the criteria from Requirement R1, R 1.2.1, a system identified in the “Applicable Systems” column. (R4)	The Responsible Entity did not notify or update E-ISAC or CISA, or their successors, within the timelines pursuant to R 4.2. (R 4.2)	The Responsible Entity did not notify E-ISAC and CISA, or their successors, of a Reportable Cyber Security Incident. (Requirement R4)

R#	Violation Risk Factor (VRF)	Violation Severity Levels (RRC-CIP-008-1)			
		Lower VSL	Moderate VSL	High VSL	Severe VSL
		The Responsible Entity notified E-ISAC and CISA, or their successors, of a Reportable Cyber Security Incident or a Cyber Security Incident that was an attempt to compromise a system identified in the “Applicable Systems” column for R 4.1 but failed to report on one or more of the attributes after determination pursuant to R 4.1. (R 4.1)			

D. Regional Variances

None.

E. Interpretations

None.

F. Associated Documents

None.

Version History

Version	Date	Action	Change Tracking

Balancing Authority	The responsible entity that integrates resource plans ahead of time, maintains Demand and resource balance within a Balancing Authority Area, and supports Interconnection frequency in real time.
BES Cyber Asset (BCA)	A Cyber Asset that if rendered unavailable, degraded, or misused would, within 15 minutes of its required operation, misoperation, or non operation, adversely impact one or more Facilities, systems, or equipment, which, if destroyed, degraded, or otherwise rendered unavailable when needed, would affect the reliable operation of the Bulk Electric System. Redundancy of affected Facilities, systems, and equipment shall not be considered when determining adverse impact. Each BES Cyber Asset is included in one or more BES Cyber Systems.
BES Cyber System (BCS)	One or more BES Cyber Assets logically grouped by a responsible entity to perform one or more reliability tasks for a functional entity.
BES Cyber System Information (BCSI)	Information about the BES Cyber System that could be used to gain unauthorized access or pose a security threat to the BES Cyber System. BES Cyber System Information does not include individual pieces of information that by themselves do not pose a threat or could not be used to allow unauthorized access to BES Cyber Systems, such as, but not limited to, device names, individual IP addresses without context, ESP names, or policy statements. Examples of BES Cyber System Information may include, but are not limited to, security procedures or security information about BES Cyber Systems, Physical Access Control Systems, and Electronic Access Control or Monitoring Systems that is not publicly available and could be used to allow unauthorized access or unauthorized distribution; collections of network addresses; and network topology of the BES Cyber System.
Blackstart Resource	A generating unit(s) and its associated set of equipment which has the ability to be started without support from the System or is designed to remain energized without connection to the remainder of the System, with the ability to energize a bus, meeting the Transmission Operator's restoration plan needs for Real and Reactive Power capability, frequency and voltage control, and that has been included in the Transmission Operator's restoration plan.
Bulk Electric System (BES)	Unless modified by the lists shown below, all Transmission Elements operated at 100 kV or higher and Real Power and Reactive Power resources connected at 100 kV or higher. This does not include facilities used in the local distribution of electric energy. Inclusions: • I1 - Transformers with the primary terminal and at least one secondary terminal operated at 100 kV or higher unless excluded by application of Exclusion E1 or E3. • I2 – Generating resource(s) including the generator terminals through the

	high-side of the step-up transformer(s) connected at a voltage of 100 kV or above with: a) Gross individual nameplate rating greater than 20 MVA. Or, b) Gross plant/facility aggregate nameplate rating greater than 75 MVA. • I3 - Blackstart Resources identified in the Transmission Operator's restoration plan. • I4 - Dispersed power producing resources that aggregate to a total capacity greater than 75 MVA (gross nameplate rating), and that are connected through a system designed primarily for delivering such capacity to a common point of connection at a voltage of 100 kV or above. Thus, the facilities designated as BES are: a) The individual resources, and The system designed primarily for delivering capacity from the point where those resources aggregate to greater than 75 MVA to a common point of connection at a voltage of 100 kV or above. • I5 - Static or dynamic devices (excluding generators) dedicated to supplying or absorbing Reactive Power that are connected at 100 kV or higher, or through a dedicated transformer with a high-side voltage of 100 kV or higher, or through a transformer that is designated in Inclusion I1 unless excluded by application of Exclusion E4 Exclusions: • E1 - Radial systems: A group of contiguous transmission Elements that emanates from a single point of connection of 100 kV or higher and: a) Only serves Load. Or, b) Only includes generation resources, not identified in Inclusions I2, I3, or I4, with an aggregate capacity less than or equal to 75 MVA (gross nameplate rating). Or, c) Where the radial system serves Load and includes generation resources, not identified in Inclusions I2, I3 or I4, with an aggregate capacity of non- retail generation less than or equal to 75 MVA (gross nameplate rating). Note 1 – A normally open switching device between radial systems, as depicted on prints or one-line diagrams for example, does not affect this exclusion. Note 2 – The presence of a contiguous loop, operated at a voltage level of 50 kV or less, between configurations being considered as radial systems, does not affect this exclusion. • E2 - A generating unit or multiple generating units on the customer's side of the retail meter that serve all or part of the retail Load with electric energy if: (i) the net capacity provided to the BES does not exceed 75 MVA, and (ii) standby, back-up, and maintenance power services are provided to the generating unit or multiple generating units or to the retail Load by a Balancing Authority, or provided pursuant to a binding obligation with a Generator Owner or Generator Operator, or under terms approved by the applicable regulatory authority. • E3 - Local networks (LN): A group of contiguous transmission Elements operated at less than 300 kV that distribute power to Load rather than transfer bulk power across the interconnected system. LN's emanate from
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	multiple points of connection at 100 kV or higher to improve the level of service to retail customers and not to accommodate bulk power transfer across the interconnected system. The LN is characterized by all of the following: a) Limits on connected generation: The LN and its underlying Elements do not include generation resources identified in Inclusions I2, I3, or I4 and do not have an aggregate capacity of non-retail generation greater than 75 MVA (gross nameplate rating); b) Real Power flows only into the LN and the LN does not transfer energy originating outside the LN for delivery through the LN; and c) Not part of a Flowgate or transfer path: The LN does not contain any part of a permanent Flowgate in the Eastern Interconnection, a major transfer path within the Western Interconnection, or a comparable monitored Facility in the ERCOT or Quebec Interconnections, and is not a monitored Facility included in an Interconnection Reliability Operating Limit (IROL). • E4 – Reactive Power devices installed for the sole benefit of a retail customer(s). Note - Elements may be included or excluded on a case-by-case basis through b) the Rules of Procedure exception process.
CIP Exceptional Circumstance	A situation that involves or threatens to involve one or more of the following, or similar, conditions that impact safety or BES reliability: a risk of injury or death; a natural disaster; civil unrest; an imminent or existing hardware, software, or equipment failure; a Cyber Security Incident requiring emergency assistance; a response by emergency services; the enactment of a mutual assistance agreement; or an impediment of large scale workforce availability.
CIP Senior Manager	A single official with the authority to bind the registered entity to adhere to the requirements within the CIP Standards, CIP-002 through CIP-015.
Control Center	One or more facilities hosting operating personnel that monitor and control the Bulk Electric System (BES) in real-time to perform the reliability tasks, including their associated data centers, of: 1) a Reliability Coordinator, 2) a Balancing Authority, 3) a Transmission Operator for transmission Facilities at two or more locations, or 4) a Generator Operator for generation Facilities at two or more locations.
Cranking Path	A portion of the electric system that can be isolated and then energized to deliver electric power from a generation source to enable the startup of one or more other generating units.
Cyber Assets	Programmable electronic devices, including the hardware, software, and data in those devices.
Cyber Security Incident	A malicious act or suspicious event that: • For a high or medium impact BES Cyber System, compromises or attempts to compromise (1) an Electronic Security Perimeter, (2) a Physical Security Perimeter, or (3) an Electronic Access Control or Monitoring System; or Disrupts or attempts to disrupt the operation of a BES Cyber System

Glossary

Distribution Provider	Provides and operates the “wires” between the transmission system and the end-use customer. For those end-use customers who are served at transmission voltages, the Transmission Owner also serves as the Distribution Provider. Thus, the Distribution Provider is not defined by a specific voltage, but rather as performing the distribution function at any voltage
Electronic Access Control or Monitoring Systems (EACMS)	Cyber Assets that perform electronic access control or electronic access monitoring of the Electronic Security Perimeter(s) or BES Cyber Systems. This includes Intermediate Systems.
Electronic Access Point	A Cyber Asset interface on an Electronic Security Perimeter that allows routable communication between Cyber Assets outside an Electronic Security Perimeter and Cyber Assets inside an Electronic Security Perimeter.
Electronic Security Perimeter	The logical border surrounding a network to which BES Cyber Systems are connected using a routable protocol.
Element	Any electrical device with terminals that may be connected to other electrical devices such as a generator, transformer, circuit breaker, bus section, or transmission line. An Element may be comprised of one or more components.
External Routable Connectivity	Programmable electronic devices, excluding Shared Cyber Infrastructure, including the hardware, software, and data in those devices. Application containers are considered software of Virtual Cyber Assets (VCAs) or Cyber Assets. VCAs are not considered software or data of Cyber Assets.
Facility	A set of electrical equipment that operates as a single Bulk Electric System Element (e.g., a line, a generator, a shunt compensator, transformer, etc.)
Interactive Remote Access	User-initiated access by a person employing a remote access client or other remote access technology using a routable protocol. Remote access originates from a Cyber Asset that is not an Intermediate System and not located within any of the Responsible Entity’s Electronic Security Perimeter(s) or at a defined Electronic Access Point (EAP). Remote access may be initiated from: 1) Cyber Assets used or owned by the Responsible Entity, 2) Cyber Assets used or owned by employees, and 3) Cyber Assets
Physical Access Control Systems	Cyber Assets that control, alert, or log access to the Physical Security Perimeter(s), exclusive of locally mounted hardware or devices at the Physical Security Perimeter such as motion sensors, electronic lock control mechanisms, and badge readers.
Physical Security Perimeter	The physical border surrounding locations in which BES Cyber Assets, BES Cyber Systems, or Electronic Access Control or Monitoring Systems reside, and for which access is controlled.

Protected Cyber Assets	One or more Cyber Assets connected using a routable protocol within or on an Electronic Security Perimeter that is not part of the highest impact BES Cyber System within the same Electronic Security Perimeter. The impact rating of Protected Cyber Assets is equal to the highest rated BES Cyber System in the same ESP.
Shared Cyber Infrastructure	One or more programmable electronic devices, including the software that shares the devices' resources, that: • Hosts one or more Virtual Cyber Assets (VCA) included in a BES Cyber Systems (BCS) or their associated Electronic Access Control or Monitoring Systems (EACMS) or Physical Access Control Systems (PACS); and hosts one or more VCAs that are not included in, or associated with, BCS of the same impact categorization; or • Provides storage resources required for system functionality of one or more Cyber Assets or VCAs included in a BCS or their associated EACMS or PACS; and also for one or more Cyber Assets or VCAs that are not included in, or associated with, BCS of the same impact categorization. SCI does not include the supported VCAs or Cyber Assets with which it shares its resources.
Remedial Action Scheme (RAS)	The following do not individually constitute a RAS: a) Protection Systems installed for the purpose of detecting Faults on BES Elements and isolating the faulted Elements b) Schemes for automatic underfrequency load shedding (UFLS) and automatic undervoltage load shedding (UVLS) comprised of only distributed relays c) Out-of-step tripping and power swing blocking d) Automatic reclosing schemes e) Schemes applied on an Element for non-Fault conditions, such as, but not limited to, generator loss-of- field, transformer top-oil temperature, overvoltage, or overload to protect the Element against damage by removing it from service f) Controllers that switch or regulate one or more of the following: series or shunt reactive devices, flexible alternating current transmission system (FACTS) devices, phase-shifting transformers, variable-frequency transformers, or tap-changing transformers; and, that are located at and monitor quantities solely at the same station as the Element being switched or regulated g) FACTS controllers that remotely switch static shunt reactive devices located at other stations to regulate the output of a single FACTS device h) Schemes or controllers that remotely switch shunt reactors and shunt capacitors for voltage regulation that would otherwise be manually switched i) Schemes that automatically de-energize a line for a non-Fault operation when one end of the line is open j) Schemes that provide anti-islanding protection (e.g., protect load from effects of being isolated with generation that may not be capable of maintaining acceptable frequency and voltage) k) Automatic sequences that proceed when manually initiated solely by a System Operator l) Modulation of HVdc or FACTS via supplementary controls, such as angle damping or frequency damping applied to damp local or inter-area

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	m) oscillations n) Sub-synchronous resonance (SSR) protection schemes that directly detect sub-synchronous quantities (e.g., currents or torsional oscillations) o) Generator controls such as, but not limited to, automatic generation control (AGC), generation excitation [e.g. automatic voltage regulation (AVR) and power system stabilizers(PSS)], fast valving, and speed governing
Reportable Cyber Security Incident	A Cyber Security Incident that compromised or disrupted: - A BES Cyber System that performs one or more reliability tasks of a functional entity; - An Electronic Security Perimeter of a high or medium impact BES CyberSystem; or - An Electronic Access Control or Monitoring System of a high or medium impact BES Cyber System
Transient Cyber Asset	A Cyber Asset that is: 1. capable of transmitting or transferring executable code, 2. not included in a BES Cyber System, 3. not a Protected Cyber Asset (PCA) associated with high or medium impact BES Cyber Systems, and 4. directly connected (e.g., using Ethernet, serial, Universal Serial Bus, or wireless including near field or Bluetooth communication) for 30 consecutive calendar days or less to a: • BES Cyber Asset, • network within an Electronic Security Perimeter (ESP) containing high or medium impact BES Cyber Systems, or • PCA associated with high or medium impact BES Cyber Systems. Examples of Transient Cyber Assets include, but are not limited to, Cyber Assets used for data transfer, vulnerability assessment, maintenance, or troubleshooting purposes.
Transmission	An interconnected group of lines and associated equipment for the movement or transfer of electric energy between points of supply and points at which it is transformed for delivery to customers or is delivered to other electric systems.
Transmission Line	A system of structures, wires, insulators and associated hardware that carry electric energy from one point to another in an electric power system. Lines are operated at relatively high voltages varying from 69 kV up to 765 kV, and are capable of transmitting large quantities of electricity over long distances.
Transmission Planner	The entity that develops a long-term (generally one year and beyond) plan for the reliability (adequacy) of the interconnected bulk electric transmission systems within its portion of the Planning Authority area.
Transmission Operator	The entity responsible for the reliability of its “local” transmission system, and that operates or directs the operations of the transmission Facilities.

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Transmission Owner	The entity that owns and maintains transmission Facilities.
Undervoltage Load Shedding Program	An automatic load shedding program, consisting of distributed relays and controls, used to mitigate undervoltage conditions impacting the Bulk Electric System (BES), leading to voltage instability, voltage collapse, or Cascading. Centrally controlled undervoltage-based load shedding is not included.

